

List of publications

– Août 2026 –

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Papers :

1. H. Jaquard, P. O. Amblard, S. Barthelmé, and N. Tremblay. Random multi-type spanning forests for synchronization on sparse graphs. *SIAM Math. of Data Science*, 7(3), 2025.
2. R. Claveau, N. Gillet, P. O. Amblard, and C. C. Finlay. Broadband a priori temporal cross-covariances for the geomagnetic inverse problem: application to the satellite era. *Earth Planets and Space*, 75:126, 2025.
3. C. Cros, P. O. Amblard, C. Prieur, and J. DaRocha. Revisiting split covariance intersection: Correlated components and optimality. *IEEE Trans. Auto. Contr.*, 70(7):4593–40607, 2025.
4. C. Cros, P.-O. Amblard, C. Prieur, and J.-F. Da Rocha. Pseudorange Rigidity and Solvability of Co-operative GNSS Positioning. *IEEE Transactions on Control of Network Systems*, 12(1):1176–1187, Mar. 2025.
5. N. Gillet, F. Dall’Asta, P.-O. Amblard, R. Claveau, and J. Aubert. Waves in Earth’s core and geomagnetic field forecast. *Physics of the Earth and Planetary Interiors*, 357:107284, 2024.
6. S. Barthelmé, P. O. Amblard, N. Tremblay, and K. Usevitch. Gaussian process regression in the flat limit. *The Annals of Statistics*, 51(6):2471–2505, Dec 2023.
7. C. Cros, P. Amblard, C. Prieur, and J. DaRocha. Fusion of distance measurements between agents with unknown correlations. *System Control Letters*, 7:2143–2148, 2023.
8. S. Barthelmé, N. Tremblay, K. Usevitch, and P. O. Amblard. Determinantal point processes in the flat limit. *Bernoulli*, 29(2):957–983, 2023.
9. N. Tremblay, S. Barthelmé, K. Usevitch, and P. O. Amblard. Extended L-ensembles: a new representation for determinantal point processes. *Annals of Applied Probability*, 33(1):613–640, 2023.
10. A. Komaty, A.-O. Boudraa, P. Flandrin, P.-O. Amblard, and J. Astolfi. On the behavior of memd in presence of multivariate fractional gaussian noise. *IEEE Trans. on Sig. Proc.*, 69:2676–2688, 2021.
11. Y. Y. Pilavci, P. O. Amblard, S. Barthelmé, and N. Tremblay. Graph tikhonov regularization and interpolation via random spanning forests. *IEEE Trans. On Signal and Information Processing over Networks*, 7:359–374, 2021. arXiv:2011.10450
12. J. F. Coeurjolly, A. Mazoyer, and P.-O. Amblard. Monte carlo integration of non differentiable functions on $[0, 1]^i, i = 1, \dots, d$, using a single determinantal point pattern defined on $[0, 1]^d$. *Electronic Journal of Statistics*, 15(2):6228–6280, 2021.

13. L. Bosse, J. Lilensten, N. Gilet, S. Rochat, A. Delboulbé, S. Curaba, A. Roux, Y. Magnard, M. Johnsen, U. Lovhaug, and P. O. Amblard. et al... On the nightglow polarisation for space weather exploration. *J. Space Weather and Space Climate*, 10:35, 2020.
14. A. Mazoyer, J. F. Coeurjolly, and P.-O. Amblard. Projections of determinantal point processes. *Spatial Statistics*, 38:100437, 2020.
15. N. Tremblay, S. Barthelmé, and P. O. Amblard. Determinantal point processes for coresets. *JMLR* 20 , 1–70 , 2019, arXiv:1803.08700
16. S. Barthelmé, P. O. Amblard, and N. Tremblay. Asymptotic equivalence of fixed-size and varying-size determinantal point processes. *Bernoulli*, 25(4B):3555–3589, 2019.
17. S. Chlailly, C. Ren, P. O. Amblard, O. J. J. Michel, P. Comon, and C. Jutten. Information-estimation relationship in mismatched gaussian channels. *IEEE Sig. Proc. Letters*, 24(5):688 – 692, 2017.
18. A. Zebadua, P. O. Amblard, E. Moisan, and O. J. J. Michel. Compressed and quantized correlation estimators. *IEEE Transaction on Signal Processing*, 65(1):56–68, 2017.
19. P. O. Amblard. A non-parametric efficient evaluation of partial directed coherence. *Biol. Cyber.*, 109:203–214, 2015.
20. J. Manton and P. O. Amblard. A primer on reproducing kernel Hilbert spaces. *Foundations and trends in signal processing*, 8(1-2):1–126, 2015.
21. Amblard, P. O. and Michel, O. J. J. Causal conditioning and instantaneous coupling in causality graphs. *Information Sciences*, 264:279–290, 2014.
22. Lilensten, J., Barthélémy, M., P. O. Amblard, and et. al,. The thermospheric auroral red line polarization: confirmation of detection and first quantitative analysis. *Space Weather and Space Climate*, 3:A01, 2013.
23. Said, S., Amblard, P. O., and Manton, J. Stationary random fields arising from second order partial differential equations on compact lie groups. *IEEE Trans on Info. Theory*, 59(6):3497–3513, 2013.
24. Amblard, P. O. and Michel, O. J. J. The relation between granger causality and directed information theory: A review. *Entropy*, 15(1):113–143, 2013.
25. Coeurjolly, J. F., Amblard, P. O., and Achard, S. Wavelet analysis of the multivariate fractional brownian motion. *ESAIM-PS*, 17:592–604, 2013.
26. McDonnell, M., Li, F., Amblard, P. O., and Grant, A. J. Optimal sensor selection for noisy binary detection in stochastic pooling networks. *Phys. Rev. E*, 88:022118, 2013.
27. Amblard, P. O., Coeurjolly, J. F., Lavancier, F., and Philippe, A. Basic properties of the multivariate fractional Brownian motion. *Bulletin Société Mathématique de France, Séminaire et Congrès*, 28:65–87, 2012.
28. Amblard, P. O. and Michel, O. J. J. On directed information theory and granger causality graphs. *Journal of Computational Neuroscience*, 30(1):7–16, 2011.
29. Amblard, P. O. and Coeurjolly, J. F. On the identification of the multivariate fractional Brownian motion. *IEEE Trans. on Signal Processing*, 59(11):5152–5168, 2011.
30. Flandrin, P., Richard, C., Amblard, P. O., Borgnat, P., Honeine, P., Amoud, H., Ferrari, A., Xiao, J., Moghtaderi, A., and Ramirez-Cobo, P. Stationnarité relative et approches connexes. *Traitement du Signal*, 28(6):691–716, 2011.

31. McDonnell, M., Amblard, P.-O., and Stocks, N. Bio-inspired communication: Performance limits for information transmission and compression in stochastic pooling networks with binary quantizing nodes. *Journal of Computational and Theoretical Nanoscience*, 7:1–8, 2010.
32. McDonnell, M., Stocks, N., and Amblard, P. O. Communication of uncoded sensor measurements through nanoscale binary-node stochastic pooling networks. *Nano Communication Networks*, 1:209–223, 2010.
33. McDonnell, M., Amblard, P.-O., and Stocks, N. Stochastic pooling networks. *Journal of Statistical Mechanics: Theory and experiment*, (2009):P01012, 2009.
34. Zozor, S., Amblard, P.-O., and Duchêne, C. Does the eye tremor provide the hyperacuity phenomenon? *Journal of Statistical Mechanics: Theory and experiment*, (2009):P01015, 2009.
35. Decrouez, G., Amblard, P. O., Brossier, J. M., and Jones, O. Galton-watson iterated function systems. *J. Phys. A: Math. Theor.*, 42:095101, 2009.
36. Kretzschmar, M., Dudok De Wit, T., Lilensten, J., Aboudarham, J., Amblard, P.-O., and Moussaoui, S. Solar euv/fuv irradiance variations: analysis and observational strategy. *Acta geophysica*, 57:42–51, 2009.
37. Dudok De Wit, T., Kretzschmar, M., Aboudarham, J., Amblard, P.-O., and Lilensten, J. Which solar EUV proxies are best for reconstructing the solar EUV irradiance? *Adv. Space. Research*, 42(5):903–911, 2008.
38. Amblard, P.-O., Moussaoui, S., Dudok De Wit, T., Aboudarham, J., Kretzschmar, M., and Lilensten, J. The EUV Sun as the superposition of elementary Sun. *Astronomy and Astrophysics*, 493:L13–L16, 2008.
39. Lilensten, J., Moen, J., Barthélémy, M., Thissen, R., Simon, C., Lorentzen, D., Dutuit, O., Amblard, P., and Sigernes, F. Polarization in aurorae: A new dimension for space environments studies. *Geophysical Research Letters*, 35:L08804, 2008.
40. Lilensten, J., Dudok de Wit, T. D., Amblard, P. O., Kretzschmar, M., Aboudarham, L., , and Auchere, F. Recommendation for a set of solar euv lines to be monitored for aeronomy applications. *Annales Geophysicae*, 25:1299–1310, 2007.
41. Zozor, S., Amblard, P., and Duchêne, C. On pooling networks and fluctuation in suboptimal detection framework. *Fluctuation and Noise Letters*, 7:39–60, 2007.
42. Amblard, P. O. and Vignat, C. A note on bounded entropies. *Physica A: statistical physics and application*, 365:50–56, 2006.
43. Zozor, S., Brossier, J. M., and Amblard, P. O. A parametric approach to suboptimal signal detection in α -stable noise. *IEEE Trans. on Signal Processing*, 54(12):4497–4509, 2006.
44. Zozor, S. and Amblard, P. O. Noise-aided processing: revisiting dithering in a $\Sigma - \Delta$ quantizer. *IEEE Trans. on Signal Processing*, 53(8):3202–3210, 2005.
45. Borgnat, P., Amblard, P. O., and Flandrin, P. Scale invariances and lamperti transformations for stochastic processes. *Journal of Physics, A : math. gen.*, 38:2081–2101, 2005.
46. Dudok De Wit, T., Lilensten, J., Aboudarham, J., Amblard, P. O., and Kretzschmar, M. Retrieving the solar euv spectrum from a reduced set of spectral lines. *Annales Geophysicae*, 23:3055–3069, 2005.
47. Senecal, S., Amblard, P. O., and Cavazzanna, L. Particle filtering equalization method for a satellite communication channel. *Journal of Applied Signal Processing*, 15:2315–2327, 2004.

48. Zozor, S. and Amblard, P. O. Stochastic resonance in locally optimal detectors. *IEEE Trans. on Signal Processing*, 51(12):3177–3181, 2003.
49. Amblard, P. O., Brossier, J., and Moisan, E. Phase tracking : what do we gain from optimality? particle filtering versus phase-locked loops. *Signal Processing*, 83:151–167, 2003.
50. Pedersen, H., Mars, J., and Amblard, P. O. Improving surface wave group velocity measurements by energy reassignment. *Geophysics*, 68(2):677–684, 2003.
51. Zozor, S. and Amblard, P. O. On the use of stochastic resonance in detection. *Signal Processing*, 82:353–367, 2002.
52. Borgnat, P., Flandrin, P., and Amblard, P. O. Stochastic discrete scale invariance. *IEEE Signal Processing Letters*, 9:181–184, 2002.
53. Senecal, S. and Amblard, P. O. Mcmc blind equalization of a satellite communication system. *Journal of Applied Signal Processing*, pages 46–57, 2002.
54. Ravier, P. and Amblard, P. O. Wavelet packets for detecting transient signals. *Signal Processing*, 81:1909–1926, 2001.
55. Lilensten, J. and Amblard, P. O. On the fast movement of the thermosphere: An interpretation. *Annales Geophysicae*, 18:1651–1656, 2001.
56. Zozor, S. and Amblard, P. O. Stochastic resonance in discrete time nonlinear ar(1) models : erratum. *IEEE Trans. on Signal Processing*, 51(5):1107–1109, 2001.
57. Amblard, P. O. and Zozor, S. Cyclostationarity and stochastic resonance in threshold devices. *Phys. Rev. E*, 5(59):5009–5020, 1999.
58. Amblard, P. O. and Brossier, J. On the cascade in fully developed turbulence. the propagator approach versus the markovian description. *European Physical Journal B*, 12:579–582, 1999.
59. Carmillet, V., Amblard, P. O., and Jourdain, G. Detection of phase- or frequency-modulated signals in reverberation noise. *Journal of the Acoustical Society of America*, 6(105):3375–3389, 1999.
60. Zozor, S. and Amblard, P. O. Stochastic resonance in discrete time nonlinear ar(1) models. *IEEE Trans. on Sig. Proc.*, 49(1):109–120, 1999.
61. Ravier, P. and Amblard, P. O. Combining an adapted wavelet analysis with 4th order statistics for transient detection. *Signal Processing*, 70:115–128, 1998.
62. Zozor, S., Moisan, E., and Amblard, P. O. Revisiting the estimation of the mean using order statistics. *Signal Processing*, 68:155–173, 1998.
63. Amblard, P. O., Gaeta, M., and Lacoume, J. L. Complex random variables and signals. part 2 : signals. *Signal Processing*, 42(11):15–25, 1996.
64. Amblard, P. O., Gaeta, M., and Lacoume, J. L. Complex random variables and signals. part 1 : variables. *Signal Processing*, 42(11):1–13, 1996.
65. Lilensten, J. and Amblard, P. O. Time-frequency tools of signal processing for eiscat data analysis. *Ann. Geophysicae*, 14(12):1513–1525, 1996.
66. Amblard, P. O. and Brossier, J. M. Adaptive estimation of the fourth-order cumulant of a white stochastic process. *Signal Processing*, 42(1):37–43, 1995.

67. Amblard, P. O. and Lacoume, J. L. A deductive construction of third-order time-frequency distributions. *Signal Processing*, 36(3):277–286, 1994.
68. Amblard, P. O. and Moisan, E. Frequency domain volterra filters in terms of distributions. *IEEE Sig. Proc. Letters*, 1(11):169–172, 1994.

Peer-Reviewed Communications in conference :

1. P. O. Amblard. Petite histoire de $\sigma_\infty^2 = \exp \int \log \gamma(\lambda) d\lambda$ et de la prédiction à un pas. In *GRETSI*, Strasbourg, 2025.
2. P.-O. Amblard, and O. J. J. Michel. Estimation en ligne des paramètres de phase et de polarisation pour les ondes gravitationnelles. In *GRETSI 2025 - XXXème Colloque Francophone de Traitement du Signal et des Images*, Strasbourg, France, Aug. 2025.
3. T. Mauger, P.-O. Amblard, and O. J. J. Michel. Approche dictionnaire pour l’identification et la quantification des ions atomiques d’un spectre de masse issu d’une sonde atomique. In *GRETSI 2025 - XXXème Colloque Francophone de Traitement du Signal et des Images*, Strasbourg, France, Aug. 2025.
4. P. Flores, J. Flamant, P.-O. Amblard, and N. Le Bihan. Décomposition de signaux bivariés en somme d’ellipses amorties. In *GRETSI 2025 - XXXème Colloque Francophone de Traitement du Signal et des Images*, Strasbourg, France, Aug. 2025.
5. P. Flores, J. Flamant, P.-O. Amblard, and N. Le Bihan. Damped ellipse decomposition for bivariate signals. In EURASIP, editor, *SSP 2025 - 2025 IEEE Statistical Signal Processing Workshop*, Edinburgh, United Kingdom, June 2025. Yoann Altmann and Mike Davies, IEEE.
6. C. Cros, P.-O. Amblard, C. Prieur, and J.-F. Da Rocha. Split Covariance Intersection with Correlated Components for Distributed Estimation. In *Fusion 2024 - 27th International Conference on Information Fusion*, Venice, Italy, July 2024.
7. N. Le Bihan, J. Flamant, and P.-O. Amblard. The geometric phase of bivariate signals. In *EUSIPCO 2024 - 32nd European Signal Processing Conference*, Lyon, France, Aug. 2024.
8. H. Jaquard, P.-O. Amblard, S. Barthelme, and N. Tremblay. Estimators for Connection-Laplacian-Based Linear Algebra. In *GSP 2024 - 7th Graph Signal Processing Workshop*, Delft, Netherlands, June 2024.
9. C. Cros, P.-O. Amblard, C. Prieur, and J.-F. Da Rocha. Intégration de mesures de distance entre agents de corrélation inconnue : cas unidimensionnel. In *GRETSI 2023 - XXIXème Colloque Francophone de Traitement du Signal et des Images*, Grenoble, France, Aug. 2023.
10. J. Flamant, P.-O. Amblard, and N. Le Bihan. Modèles physiques à deux états pour les signaux bivariés : un système d’oscillateurs harmoniques couplés. In *GRETSI 2023 - XXIXème Colloque Francophone de Traitement du Signal et des Images*, Grenoble, France, Aug. 2023. GRETSI - Groupe de Recherche en Traitement du Signal et des Images.
11. H. Jaquard, P.-O. Amblard, S. Barthelme, and N. Tremblay. Angular Synchronization on Graphs with Monte-Carlo. In *GSP 2023 - 6th Graph Signal Processing workshop*, Oxford, United Kingdom, June 2023.
12. H. Jaquard, M. Fanuel, P.-O. Amblard, R. Bardenet, S. Barthelme, and N. Tremblay. Smoothing complex-valued signals on Graphs with Monte-Carlo. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Rhodes Island, Greece, June 2023.

13. N. Le Bihan, J. Flamant, and P.-O. Amblard. Modèles physiques à deux états pour les signaux bivariés : modulation de polarisation et phase géométrique. In *GRETSI 2023 - XXIXème Colloque Francophone de Traitement du Signal et des Images*, Grenoble, France, Aug. 2023. GRETSI - Groupe de Recherche en Traitement du Signal et des Images.
14. N. Tremblay, S. Barthelmé, and P. Amblard. A faster sampler for discrete determinantal point processes. In *AISTAT, PMLR*, 2023.
15. N. Tremblay, Y. Y. Pilavci, S. Barthelme, and P.-O. Amblard. What can we Compute With Kirchhoff Forests? In *GSP 2023 - 6th Graph Signal Processing workshop*, Oxford, United Kingdom, June 2023.
16. C. Cros, C. Prieur, P.-O. Amblard, and J.-F. da Rocha. Résolubilité du positionnement GNSS multi-agents. In *GRETSI 2022 - XXVIIIème Colloque Francophone de Traitement du Signal et des Images*, Nancy, France, Sept. 2022.
17. Y. Y. Pilavci, P.-O. Amblard, S. Barthelme, and N. Tremblay. Variance Reduction for Inverse Trace Estimation via Random Spanning Forests. In *GRETSI 2022 - XXVIIIème Colloque Francophone de Traitement du Signal et des Images*, Nancy, France, Sept. 2022.
18. P.-O. Amblard, S. Barthelme, N. Tremblay, and K. Usevich. Mesures d'indépendance dans des rkHs en limite plate. In *GRETSI 2022 - XXVIIIème Colloque Francophone de Traitement du Signal et des Images*, Nancy, France, Sept. 2022.
19. Y. Y. Pilavci, P. O. Amblard, S. Barthelmé, and N. Tremblay. Smoothing graph signals via random spanning forests. In *proc ICASSP, Barcelona, Spain*, 2020.
20. S. Barthelmé, N. Tremblay, A. Gaudillière, L. Avena, and P. O. Amblard. Estimating the inverse trace using random forests. In *GRETSI 2019, Lille, France*, 2019.
21. P. O. Amblard, S. Barthelmé, and N. Tremblay. Subsampling with k determinantal point processes for estimating statistics in large data sets. In *IEEE SSP, Freiburg, Germany*, 2018.
22. A. Zebadua, E. Moisan, and P. O. Amblard. Sur le corrélateur compressé : Estimation de temps de retard et d'interspectre. In *Proc. GRETSI, Juans-Les-Pins*, 2017.
23. P. O. Amblard and J. F. Coeurjolly. sous-échantillonnage aléatoire par processus déterminantaux pour l'estimation. In *Proc. GRETSI*, 2017, Nice.
24. N. Tremblay, P. O. Amblard, and S. Barthelmé. Graph sampling with determinantal processes. In *EUSIPCO, Kos Island, Greece, arXiv 1703.01594*, 2017.
25. C. Ren, R. C. Farias, P.-O. Amblard, and P. Comon. Performances optimales pour les données multi-modales partiellement couplées. In *GRETSI 2017, Juans-Les-Pins, France*, 2017.
26. S. Chlailly, P.-O. Amblard, O. J. J. Michel, and C. Jutten. L'interaction entre deux modalités complémentaires. In *GRETSI 2017, Juans-Les-Pins, France*, 2017.
27. A. Bourrier, P. O. Amblard, O. J. J. Michel, and C. Jutten. Multimodal kalman filter. In *ICASSP, Shanghai, China*, 2016.
28. S. Chlailly, P. O. Amblard, O. J. J. Michel, and C. Jutten. Impact of noise correlation on multimodality. In *Proc. EUSIPCO, Budapest, Hongrie*, 2016.
29. C. Ren, R. C. Farias, P. O. Amblard, and P. Comon. Performance bounds for coupled models. In *IEEE SAM, Brazil*, 2016.

30. G. Decrouez and P. O. Amblard. Analysis of h-sssi processes using the crossing tree: an alternative to wavelets. In *proc ICASSP, Brisbane, Australia*, 2015.
31. G. Decrouez and P. O. Amblard. Crossing tree partition functions. In *Proc. EUSIPCO, Nice, France*, 2015.
32. P. O. Amblard and H. Kadri. Algorithme lms pour la régression à noyau multivariée parcimonieuse. In *GRETSI 2015, Lyon, France*, 2015.
33. P. O. Amblard and H. Kadri. Operator-valued kernel recursive least squares algorithm. In *Proc. EUSIPCO, Nice, France*, 2015.
34. A. Zebadua, P. O. Amblard, E. M. O. J. J. Michel, M. Carmona, A. Paleologue, and M. Billere. Examples of output-only modal identification using compressive sensing techniques. In *IWSHM 2015, Stanford, USA*, 2015.
35. A. Zebadua, P. O. Amblard, E. Moisan, and O. J. J. Michel. Estimation de corrélation sur signaux compressés. In *GRETSI 2015, Lyon, France*, 2015.
36. A. Zebadua, P.-O. Amblard, O. J. J. Michel, and E. Moisan. Quantization and compressive sensing for correlation estimation in sensor networks. In *IMA Conf. on Maths and Signal Processing, Birmingham*, 2014.
37. Amblard, P. O. and Michel, O. J. J. Granger causality for functional valued random processes. In *International Symposium on Information Theory and its Applications*. Melbourne, 2014.
38. P. O. Amblard. Versions récursives et adaptatives d’une mesure d’indépendance. In *GRETSI 2013, Brest, France*, 2013.
39. P. O. Amblard and O. J. J. Michel. Causalité de granger pour les signaux à valeurs fonctionnelles. In *GRETSI 2013, Brest, France*, 2013.
40. P. O. Amblard and J. H. Manton. On data sparsification and a recursive algorithm for estimating a kernel-based measure of independence. In *proc. ICASSP, Vancouver, Canada*, 2013.
41. G. Decrouez, P. O. Amblard, and O. Jones. Estimation of the multifractal spectrum using the crossing tree. In *GRETSI 2013, Brest, France*, 2013.
42. P. O. Amblard and O. J. J. Michel. Causal conditioning and instantaneous coupling in causality graphs. In *IEEE SSP, Ann Arbor, USA*, 2012.
43. P.-O. Amblard, O. J. J. Michel, C. Richard, and P. Honeine. A Gaussian process regression approach for testing Granger causality between time series data. In *proc. ICASSP, Osaka, Japan*, 2012.
44. P. O. Amblard, R. Vincent, O. J. J. Michel, and C. Richard. Kernelizing geweke’s measure of granger causality. In *IEEE workshop on MLSP, Santander, Spain*, sept. 2012.
45. J. Lilensten, M. Barthélémy, and P.-O. Amblard. et al. Calibrated polarization parameters of the auroral red line in hornsund. In *conference EGU, Vienna, Austria*, 2012.
46. S. Said, J. Manton, and P.-O. Amblard. On a class of filtering problems with observation in a riemannian manifold. In *proc. MTSN, Melbourne, Australia*, 2012.
47. P. O. Amblard and J. F. Coeurjolly. Identification du mouvement brownien fractionnaire multivarié. In *proc. Grets, Bordeaux, France*, page CDROM, 2011.

48. P. O. Amblard and O. J. J. Michel. Information dirigée et graphes de causalité au sens de granger. In *proc. Grets, Bordeaux, France*, page CDROM, 2011.
49. F. Chatelain, S. Achard, C. Gouy-Pailler, O. J. J. Michel, and P. O. Amblard. Graphe de connectivité cérébrale et longue dépendance. In *proc. Grets, Bordeaux, France*, 2011.
50. P. O. Amblard and O. J. J. Michel. Directed information theory to infer causality graphs. In *Statworks, Statistics and Inference for Networks*, 2010.
51. J. F. Coeurjolly, P. O. Amblard, and S. Achard. On multivariate fractional brownian motion and multivariate fractional gaussian noise. In *in Proceedings EUSIPCO 2010, Aalborg, Denmark*, 2010.
52. F. Li, M. McDonnell, P. O. Amblard, and A. J. Grant. Sensor selection for distributed detection via multiaccess channels. In *Proc. IEEE Australian Communication Theory Workshop, Sydney*, pp77–82, Feb 2009, Best Student Paper Award.
53. P. O. Amblard, S. Achard, J. Lemoine, and C. Gouy-Pailler. Retrieving connectivity in complex networks of long range dependent stochastic processes. In *European Conference on Complex Systems, CD ROM, Track F - Mathematics and Simulation*, p76, 2009.
54. P. O. Amblard and O. J. J. Michel. Sur différentes mesures de dépendance causale entre signaux aléatoires (on different measures of causal dependence for stochastic processes). In *proc. Grets, Dijon, France, Sept.*, 2009.
55. J. Lemoine, C. Gouy-Pailler, S. Achard, and P. O. Amblard. Recherche de la connectivité de réseaux complexes. application en irmf. In *proc. Grets, Dijon, France, Sept.*, 2009.
56. P. O. Amblard and O. J. J. Michel. Information flow through scales. In *IMA conf on Maths and Signal processing*, pp78, 2008.
57. P. O. Amblard, S. Zozor, O. J. J. Michel, and A. Cuculescu. On the estimation of the entropy using k -th nearest neighbors. In *IMA Conf. on Maths and Signal Processing*, pp 79-82, 2008.
58. G. Becq, P. O. Amblard, S. Roth, and C. Villard. Simulation of hippocampal neural networks under bicuculline, with noise and synaptic delays. In *Neurocomp 2008*, 2008.
59. E. Diaz, R. Escola, P. Pham, R. Guillemaud, G. Becq, P.-O. Amblard, and C. Villard. Recording and simulation of hippocampal neural networks. In *MEA 2008*, 2008.
60. P. O. Amblard, S. Moussaoui, T. Dudok de Wit, J. Liliensten, M. Kretschmar, J. Aboudarham, and F. Auchere. Le soleil comme superposition de soleils élémentaires. In *GRETSI 2007, Troyes, France*, 2007.
61. P. O. Amblard, S. Zozor, and M. McDonnell. Pooling networks for a discrimination task: noise-enhanced detection. In *4rd SPIE conf. on Noise and Fluctuations, Firenze, Italia, vol. , Invited Talk*, 2007.
62. G. Decrouez, P. O. Amblard, J. M. Brossier, and O. Jones. Galton-watson random fractals. 2007.
63. G. Decrouez, P. O. Amblard, J. M. Brossier, and O. Jones. Systèmes de fonctions itérés aléatoires sur des arbres de galton-watson. In *GRETSI 2007, Troyes, France*, 2007.
64. C. Villard, P. Amblard, G. Becq, S. Gory-Faure, and S. Roth. Living ordered neural networks as model systems for signal processing. In *4rd SPIE conf. on Noise and Fluctuations, Firenze, Italia, CDROM , Invited Talk*, 2007.

65. S. Zozor, P.-O. Amblard, and C. Duchêne. Echantillonnage aléatoire de la rétine et micro-saccades: amélioration du traitement de l'information par le bruit ? In *GRETSI 2007, Troyes, France*, 2007.
66. S. Zozor, P.-O. Amblard, and C. Duchêne. Fluctuation in the retina: noise-enhanced processing via random sampling and microsaccads. In *4rd SPIE conf. on Noise and Fluctuations, Firenze, Italia, vol. ,*, 2007.
67. S. Zozor, P.-O. Amblard, and C. Duchêne. On the microsaccades in the retina as a noise-enhanced processing. In *proc. Physics in Signal and Image Processing, PSIP 2007 Mulhouse, France*, 2007.
68. P. O. Amblard, O. J. J. Michel, and S. Morfu. Detection and information points of view in threshold devices. In *proc. Nonlinear Dynamical Electronic Sysems, NDES 2006, Dijon, France*, pages pp1–4, June 2006.
69. P. O. Amblard, S. Zozor, and O. J. J. Michel. Networks of the pooling type and optimal quantization. In *proc. IEEE ICASSP 2006, CDROM, Toulouse, France*, 2006.
70. N. L. Bihan and P. O. Amblard. Detection and estimation of gaussian proper quaternion valued random processes. In *proc. IMA conf. on mathematics in signal processing, Cirencester, UK*, pages 23–26, 2006.
71. G. Decrouez, P. O. Amblard, and J. M. Brossier. A wavelet analysis of random iterated function systems. In *proc. IMA conf. on mathematics in signal processing, Cirencester, UK*, pages 14–17, 2006.
72. S. Gory-Faure, J. Brocard, P. Amblard, A. Depaulis, P. Salin, E. Dumas, S. Roth, and C. Villard. Ordered neural networks grown on patterned mea. In *proc. MEA Meeting, Reutigen, Germany*, pages pp194–195, July 2006.
73. M. McDonnell, P. O. Amblard, N. Stocks, S. Zozor, and D. Abbott. High resolution optimal quantization for stochastic pooling networks. In *SPIE conference on Complexity and nonlinear dynamics systems, vol. 6417, Invited Talk*, pages 100–110, Dec. 2006.
74. P. O. Amblard, O. J. J. Michel, and S. Morfu. Revisiting the asymmetric binary channel: joint noise-enhanced detection and information transmission through threshold devices. In *3rd SPIE conf. on Noise and Fluctuations, Austin, USA, vol. 5845*, pages 50–60, May 2005.
75. P. O. Amblard and C. Vignat. A note on bounded entropies. In *NEXT2005, News, EXpectations and Trends in Statistical Physics*, pages –, 2005.
76. C. Duchêne, P.-O. Amblard, and S. Zozor. Quantizer based suboptimal detectors: noise-enhanced performance and robustness. In *3rd SPIE conf. on Noise and Fluctuations, Austin, USA, vol. 5847, Invited Talk*, pages 100–110, May 2005.
77. C. Duchêne, S. Zozor, and P.-O. Amblard. Détection sous-optimale et quantifieurs stochastiques. In *proc. GRETSI 2005, Louvain-La-Neuve, Belgique*, page CDROM, 2005.
78. C. Duchêne, S. Zozor, and P.-O. Amblard. Noise-aided suboptimal quantizer-decision systems. In *proc. Physics in Signal and Image Processing, PSIP 2005, Toulouse, France*, pages 43–48, 2005.
79. A. Voicila, C. Duchêne, S. Zozor, J. Brossier, and P. O. Amblard. On a suboptimal detector in α -stable context. In *proc. Physics in Signal and Image Processing, PSIP 2005, Toulouse, France*, pages 289–294, 2005.
80. S. Zozor, J. Brossier, P. O. Amblard, C. Duchêne, and A. Voicila. About suboptimal detection in α -stable noise. In *3rd SPIE conf. on Noise and Fluctuations , Austin, USA, vol. 5847*, pages 111–120, 2005.

81. S. Zozor, J. M. Brossier, and P. O. Amblard. Détection sous-optimale dans du bruit α -stable. In *proc. GRETSI 2005, Louvain-La-Neuve, Belgique*, page CDROM, 2005.
82. P. O. Amblard and N. L. Bihan. On properness of quaternion valued random variables. In *proc. IMA conf. on mathematics in signal processing, Cirencester, UK*, pages 23–26, 2004.
83. P. O. Amblard and P. Borgnat. Stochastic discrete scale invariance : renormalization group operators and iterated function systems. In *proc. IMA conf. on mathematics in signal processing, Cirencester, UK*, pages 19–22, 2004.
84. J. M. Brossier and P. O. Amblard. Particulaire et communications numériques. In *Journées SEE Signal et communications numériques*, 2004.
85. P. O. Amblard, P. Borgnat, and P. Flandrin. Stochastic processes with finite size scale invariance. pages 507–518, 2003.
86. P. Borgnat, P. O. Amblard, and P. Flandrin. Invariance d'échelle brisée et accroissements stationnaires. In *proc. GRETSI 2003, Paris, France*, page CDROM, 2003.
87. P. Borgnat, P. O. Amblard, and P. Flandrin. Lamperti transformation for finite size scale invariance. In *proc. Physics in Signal and Image Processing, PSIP 2003, Grenoble, France*, pages 177–180, 2003.
88. S. Senecal, P. O. Amblard, and L. Cavazzanna. Sequential monte carlo method for blind equalization of a nonlinear satellite communication system. In *proc ICASSP, Hong-Kong, cancelled because of SRAS*, page cdrom, 2003.
89. S. Zozor and P. O. Amblard. On stochastic resonance-like effects in detection. pages 161–172, 2003.
90. S. Zozor and P. O. Amblard. Stochastic resonance in a sigma-delta quantizer. In *proc. IEEE workshop Nonlinear Digital Signal and Image Processing, Grado, Italy*, pages CD-ROM, 2003.
91. S. Zozor and P. O. Amblard. Stochastic resonance in terms of detection. In *proc. Physics in Signal and Image Processing, PSIP 2003, Grenoble, France*, pages 153–156, 2003.
92. P. O. Amblard, J. M. Brossier, and E. Moisan. Tracking a brownian phase with linear drift: algorithms and performances. In *Proc. EUSIPCO, Toulouse, France*, pages CD-ROM, 2002.
93. P. Borgnat, P. O. Amblard, and P. Flandrin. Generalized lamperti transformation of broken scale invariance. In *ASILOMAR conf. on Signal, Systems and Computers*, page CDROM, 2002.
94. P. Borgnat, P. O. Amblard, and P. Flandrin. Random processes with discrete scale invariance: Methods of synthesis. In *Symp. Self-similarity and Applications, Clermont-Ferrand France*, 2002.
95. J. M. Brossier, P. O. Amblard, and B. Geller. Self-adaptive PLL for general QAM constellations. In *Proc. EUSIPCO, Toulouse, France*, pages CD-ROM, 2002.
96. P. O. Amblard, J. M. Brossier, and E. Moisan. Poursuite d'une phase brownienne à dérive linéaire. performances et algorithmes. In *Proc. GRETSI, Toulouse, France*, pages 203–206, 2001.
97. P. Borgnat, P. Flandrin, and P. O. Amblard. Stochastic discrete scale invariancee. In *Proc. IEEE workshop on Stat. Signal. Proc., Singapore*, pages 66–69, 2001.
98. P. Borgnat, P. Flandrin, and P. O. Amblard. Une approche statistique de l'invariance d'échelle discrète. In *Proc. GRETSI, Toulouse, France*, pages 29–32, 2001.

99. S. Senecal and P. O. Amblard. Mcmc methods for discrete sources separation. In *Bayesian inference and maximum entropy methods in science and engineering, Proc. MaxEnt'00, Gif-sur-Yvette, France*, pages 350–360, 2001.
100. S. Senecal and P. O. Amblard. Mcmc methods non linear channel equalization. In *Proc. IEEE NSIP, USA*, page CD Rom, 2001.
101. S. Senecal and P. O. Amblard. Méthodes de simulation pour la séparation de sources discrètes. In *Proc. GRETSI, Toulouse, France*, pages 319–322, 2001.
102. S. Zozor and P. O. Amblard. Les détecteurs localement optimaux sont -ils des résonateurs stochastiques. In *Proc. GRETSI, Toulouse, France*, pages 703–706, 2001.
103. S. Senecal and P. O. Amblard. Bayesian separation of discrete sources *via* Gibbs sampling. In *Proc. ICA, Helsinki, Finland*, pages 561–566, 2000.
104. S. Zozor and P. O. Amblard. Can stochastic resonance be used in detection. In *Proc. EUSIPCO, Tampere, Finland*, 2000.
105. P. O. Amblard and O. J. J. Michel. On higher-order affine distributions. In *Proc. IEEE workshop on HOS, Haifa Israel* pages 273–276, 1999.
106. P. O. Amblard and O. J. J. Michel. Vers des représentations affines multilinéaires. In *Proc. colloque GRETSI, Vannes, Tome 1*, pages 19–22, 1999.
107. J. Mars, P. O. Amblard, H. Pedersen, and A. L. Bard. Réallocation énergétique pour l'analyse par filtrage multiple. application à la mesure de courbe de dispersion de séismes. In *Proc. colloque GRETSI, Vannes, Tome 2*, pages 343–346, 1999.
108. S. Zozor and P. O. Amblard. 2-dimensional stochastic resonance in a discrete time nonlinear system with an elliptic input. In *Proc. IEEE workshop on nonlinear signal processing, Antalya, Turquie*, pages 95–99, jun 1999.
109. S. Zozor and P. O. Amblard. Qu'en-est-il de la résonance stochastique comme outil potentiel en traitement du signal. In *Proc. colloque GRETSI, Vannes, Tome 2*, pages 451–454, 1999.
110. S. Zozor and P. O. Amblard. Stochastic resonance in a discrete time nonlinear system with cyclostationarity inputs. In *Proc. Physics in Signal and Image Processing, Paris, France*, pages –, 1999.
111. P. Ravier and P. O. Amblard. Denoising using wavelet packets and the kurtosis: Application to transient detection. In *Proc. IEEE symposium on time-frequency and time-scale analysis, Pittsburgh, USA*, pages 625–628, 1998.
112. P. O. Amblard, J. M. Brossier, and J. L. Lacoume. Playing with long range dependence and hos. In *Proc. IEEE workshop on HOS, Banff, Canada*, pages 453–457, 1997.
113. S. Zozor and P. O. Amblard. Résonance stochastique via des setar $(1,n, 0,...,0)$ discrets non linéaires. In *Proc. GRETSI, Grenoble*, pages 773–776, 1997.
114. S. Zozor and P. O. Amblard. Stochastic resonance in a discrete time nonlinear setar $(1,2,0,0)$ model. In *Proc. IEEE workshop on HOS, Banff, Canada*, pages 166–170, 1997.
115. S. Zozor, E. Moisan, and P. O. Amblard. Estimation de la moyenne et statistiques d'ordre. In *Proc. GRETSI, Grenoble*, pages 507–510, 1997.

116. V. Pierson and P. O. Amblard. Studying the performances of linear signature detector on a time-frequency representation. In *Proc. IEEE digital signal processing*, September 1996.
117. P. Ravier and P. O. Amblard. A transient detector based on malvar wavelets. In *IEEE Ocean, Fort Lauderdale, USA*, September 1996.
118. P. Ravier and P. O. Amblard. Using malvar wavelets for transient detection. In *IEEE symposium on time-frequency and time-scale analysis, Paris, France*, pages 229–232, June 1996.
119. P. O. Amblard and P. Duvaut. Filtrage adapté dans le cas gaussien complexe non circulaire. In *Proc. GRETSI, Juans-Les-Pins*, pages 141–144, 1995.
120. P. O. Amblard and E. Moisan. On the use of frequency domain volterra filters. In *IEEE Workshop on nonlinear signal and image processing*, pages 380–383, Halkidiki, Greece, 20–22 June 1995.
121. P. O. Amblard and P. Ravier. Experimental performance analysis of an on-line transient detector. In *IEEE Workshop on HOS*, pages 171–175, Gerona, Spain, 12–14 June 1995.
122. L. Duboisset, P. Ravier, and P. O. Amblard. Etudes des performances de détecteurs de transitoires fondés sur les statistiques d’ordre supérieurs et les transformées linéaires. In *Proc. GRETSI, Juans-Les-Pins*, pages 1169–1171, 1995.
123. P. Marchand, P. O. Amblard, and J. L. Lacoume. Statistiques d’ordre supérieur à deux pour des signaux cyclostationnaires à valeurs complexes. In *Proc. GRETSI, Juans-Les-Pins*, pages 69–72, 1995.
124. K. Mio, E. Moisan, and P. O. Amblard. Nonlinear noise cancellation: Bilinear filtering of a sonar experiment. In *IEEE Workshop on nonlinear signal and image processing*, pages 416–419, Halkidiki, Greece, 20–22 June 1995.
125. P. O. Amblard and D. Baudois. Non-linear noise cancellation using volterra filters : a real case study. In *IEEE workshop on Nonlinear Digital Signal Processing*, pages 2.1–1.1–1.5, Tampere, Finland, January 1993.
126. P. O. Amblard, J. M. Brossier, and N. Charkani. New adaptive estimation of the fourth-order cumulant... In *Proc. EUSIPCO*, pages 466–469, Edinburgh, Sept. 1994.
127. P. O. Amblard and D. Baudois. Non-linear noise cancellation using volterra filters : a real case study. In *IEEE workshop on Nonlinear Digital Signal Processing*, pages 2.1–1.1–1.5, Tampere, Finland, January 1993.
128. P. O. Amblard and J. L. Lacoume. Statistiques d’ordre supérieur des signaux complexes stationnaires et non-stationnaires. In *Proc. GRETSI*, pages 121–124, 1993.
129. P. O. Amblard, J. L. Lacoume, and J. M. Brossier. Transient detection, higher-order time-frequency distributions and the entropy. In *Proc. IEEE SP Workshop on Higher-Order Stat., Lake Tahoe, USA*, pages 265–269, 1993.
130. P. O. Amblard, D. Baudois, and J. L. Lacoume. Edgeworth series expansion of the conditional mean and the optimality of non-linear volterra filters. In *Proc. ICASSP*, pages V477–V480, San Francisco, March 1992.
131. P. O. Amblard and J. Lacoume. Construction of fourth-order cohen’s class : a deductive approach. In *IEEE symposium on Time-Scale and Time-Frequency Analysis*, Victoria, Canada, October 1992.
132. J. L. Lacoume, M. Gaeta, and P. O. Amblard. From order 2 to HOS: new tools and applications. In *Proc. European Signal Processing Conf. EUSIPCO*, pages 91–98, Brussels, Aug 24 - 27 1992.

133. P.-O. Amblard, D. Baudois, and C. Servière. Soustraction de bruit en situation non linéaire utilisant les filtres de volterra. In *Proc. GRETSI*, pages 325–328, Juans-Les-Pins, Sep - 1991.

Invited conferences :

1. P.O. Amblard (et équipe CICS du GIPSA), Quelques aspects de traitement de l'information compressée pour les capteurs en réseaux, Journées Objets Communicants : Algorithmes, Architectures et Applications, organisées par l'INS2I, CNRS, Paris, juillet 2017
2. P.-O. Amblard C. Jutten, S. Chlailly and O. Michel. Effects of model mismatches in multimodal estimations. In SBrT 2017, San Pedro, Brazil, September 2017.
3. P.O. Amblard, On the links between directed information theory and Granger causality Workshop Information Processing and Cognition, Sydney, Australia, 27-28 Feb 2013
4. P.O. Amblard Granger causality and directed information theory, invited conference at Workshop on directed information theory, Munich, May 2012.
5. P.O. Amblard and O.J.J. Michel, Networks of signals, directed information and causality graphs, Invited Seminar, SEMOVI, GIN, Grenoble, France, 7 décembre 2011
6. Jean Lilensten Mathieu Barthélemy, Roland Thissen, J-ran Moen, Cyril Simon, Dag A. Lorentzen, Odile Dutuit, Pierre Olivier Amblard, Fred Sigernes Polarization in aurorae: a new dimension for space environments studies invited conference at the Space Research Centre Polish Academy of Sciences, april 2008, Warsaw, Poland
7. P.O. Amblard, S. Zozor, M. McDonnell and N. Stocks Pooling networks for a discrimination task: noise-enhanced detection. In *4rd SPIE conf. on Noise and Fluctuations, Firenze, Italia, vol. , Invited Talk*, 2007.
8. C. Villard, P.O. Amblard, G. Becq S. Gory-Fauré, and S. Roth Living ordered neural networks as model systems for signal processing. In *4rd SPIE conf. on Noise and Fluctuations, Firenze, Italia, CDROM , Invited Talk*, 2007.
9. M. McDonnell, P. O. Amblard, N. Stocks, S. Zozor, and D. Abbott. High resolution optimal quantization for stochastic pooling networks. In *SPIE conference on Complexity and nonlinear dynamics systems, vol. 6417, Invited Talk*, pages 100–110, December 2006.
10. J. Lilensten, T. Dudok de Wit, M. Kretzschmar, P.O. Amblard, S. Moussaoui J. Aboudarham, F. Auchère, Retrieving the EUV solar spectrum from a selected set of lines for space weather purposes: A review of theories, models and experiments, invited conference , European Space Weather Week, nov 2006, Bruxelles
11. C. Duchêne, P. O. Amblard, and S. Zozor. Quantizer based suboptimal detectors: noise-enhanced performance and robustness. In *3rd SPIE conf. on Noise and Fluctuations, Austin, USA, vol. 5847, Invited Talk*, pages 100–110, May 2005.
12. C. Duchêne, P. O. Amblard, and S. Zozor. Quantizer based suboptimal detectors: noise-enhanced performance and robustness. In *3rd SPIE conf. on Noise and Fluctuations, Austin, USA, vol. 5847, Invited Talk*, pages 100–110, May 2005.
13. P.-O. Amblard. Le bruit peut-il devenir un allié? Traitements de l'information favorisés par le bruit, Exposition de Physique, Entretiens Physique-Industrie : Le Bruit, 17 juin 2004
14. P.-O. Amblard. Résonance stochastique Assemblée générale du GdR ISIS, Hourtin, mai 2001.

Book :

J.L. Lacoume, P.O. Amblard, and P. Comon. *Statistiques d'ordre supérieur pour le traitement du signal*. Masson, 1997.

Book chapters

1. P.O. Amblard and P. Ravier. *Détection dans le plan temps-fréquence*, chapter Détection de transitoires par ondelettes adaptées, pages 84–102. C. Doncarli, N. Martin eds., Hermès, 2004.
2. P.O. Amblard. *Temps-fréquence-Concept et outils*, chapter Représentations temps-fréquence d'ordres supérieurs, pages 239–258. F. Hlawatsch, C. Auger eds., Hermès, 2005.
3. P. O. Amblard. *Time-frequency analysis (translation of Temps-fréquence-Concept et outils)*, chapter Higher-order time-frequency representations. F. Hlawatsch, C. Auger eds., ISTE, 2006.
4. P. Flandrin, P. Borgnat, and P.O. Amblard. *Long-Range Dependent Stochastic Processes: Theory and Applications*, chapter From stationarity to self-similarity and back: Variations on the Lamperti transform. G. Rangajanan ed., Springer, 2003.

Thesis :

P.O. Amblard. *Statistiques d'ordre supérieur pour les signaux non gaussiens, non linéaires, non stationnaires*. Thèse de Doctorat de l'INPG, 1994.

Habilitation à Diriger des Recherches :

P.O. Amblard. *Non, non et non, ou quelques études sur et autour des non-propriétés en traitement du signal*. mémoire d'Habilitation à Diriger des Recherches, INPG, 2001.

Summer School :

1. J.M. Brossier and P.-O. Amblard. Introduction aux problèmes de synchronisation Ecole d'été ECOTEL, Nice, octobre 2000